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Cold Chain Logistics Performance in Saudi Arabia: Evidence from the Pharmaceutical Industry

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Abstract

Cold chain logistics play a key role in making sure that pharmaceutical products stay safe and effective from the moment they are stored until they reach patients. This is especially important in Saudi Arabia, where very high temperatures can make it difficult to maintain the required conditions. Because of this, understanding how well cold chain systems actually work in practice is essential. This study looks at the effectiveness of cold chain logistics in the Saudi pharmaceutical sector using real life case studies. Two organizations: Tamer Group and Medical Fakeeh were selected to represent different parts of the supply chain. This helped provide a clearer and more complete picture of how these systems operate. Data were collected through interviews with professionals working directly in logistics and healthcare, along with supporting operational data such as temperature records. The findings show that cold chain practices are generally reliable and supported by modern technologies like data loggers and monitoring systems. However, challenges still exist, especially due to extreme heat, day-to-day operational issues, and human error. The study also shows that companies use different approaches; some focus on flexibility and quick response, while others focus more on strict control and compliance. Overall, the study suggests that while current systems are effective, there is still a need for continuous improvement to ensure consistent performance and patient safety.

Keywords
Cold Chain Logistics, Pharmaceutical Supply Chain, Temperature Control, Logistics Performance Cold Chain Systems in Saudi Arabia.

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1. Introduction

1.1 Background on Cold Chain Logistics

Cold chain logistics refers to a system where products are stored and transported under controlled temperature conditions to keep them safe and effective. This is important, especially in the pharmaceutical industry, where products like vaccines, insulin, and tablets are highly sensitive to temperature changes. If these products are exposed to temperatures outside their range, the quality can be affected, making them unsafe to use, so this can lead to financial losses and, more importantly, risks to patient health. For this reason, maintaining a reliable cold chain throughout the entire supply process is important to ensure that pharmaceutical products reach the patient in proper conditions

1.2 Importance in Saudi Arabia

Cold chain logistics are important in Saudi Arabia because of the country's very hot climate. High temperatures, especially during the summer, make it more difficult to keep pharmaceutical products within the required temperature range during storage and transportation. In addition, the long distances between cities can increase the chances of temperature changes along the supply chain. At the same time, the demand for pharmaceutical products in Saudi Arabia is growing due to population growth and ongoing improvements in the healthcare sector. With initiatives like Vision 2030, there is a strong focus on enhancing healthcare services, which makes reliable cold chain logistics even more important. Ensuring proper temperature control is therefore essential to guarantee that medicines remain safe and effective when they reach patients.

1.3 Problem statement

Even though cold chain logistics play a very important role in the pharmaceutical sector, there is still limited real evidence showing how effective these practices are in Saudi Arabia. The country faces several challenges that can affect the performance of the cold chain, such as extreme weather conditions, long transportation distances, and day-to-day operational issues. For example, temperature changes can happen during the final stage of delivery, known as last-mile delivery, where control is often more difficult. In addition, human errors in handling and the high cost of advanced monitoring technologies can make it harder for companies to maintain consistent standards. Because of these challenges, it is important to better understand how well cold chain systems are currently working and where improvements are needed to ensure product quality and patient safety.

1.4 Research Question

How effective are cold chain logistics practices in Saudi Arabia's pharmaceutical sector?

1.5 Research Objectives

The study aims to

- Understand how effective cold chain logistics practices are in Saudi Arabia's pharmaceutical sector
- Evaluate whether temperature-controlled systems are properly maintained throughout the supply chain
- Identify the main challenges affecting performance, such as climate conditions, operational issues, and human factors
- Examine the role of monitoring technologies in supporting cold chain operations
- Suggest practical improvements to ensure product quality and patient safety

1.6 Case Study basis

This research is based on case studies of two major organizations in Saudi Arabia's healthcare and pharmaceutical logistics sector: **Tamer Group** and **Medical Fakeeh (Saudi Airlines Medical Services)**. These companies play a key role in pharmaceutical distribution and healthcare services, making them suitable examples for evaluating real-world cold chain logistics practices.

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2. Literature Review

Cold chain logistics refers to a temperature-controlled supply chain that ensures the safe storage, handling, and transportation of temperature-sensitive products. In the pharmaceutical sector, maintaining strict temperature conditions is essential, as products such as vaccines, biologics, and insulin are highly sensitive to environmental changes. Even minor temperature deviations can reduce drug effectiveness, create financial losses, and pose serious risks to patient safety. Effective cold chain systems are therefore critical for preserving product integrity and ensuring reliable healthcare delivery.

Managing cold chain logistics in the pharmaceutical industry is particularly complex due to strict regulatory requirements and the need for continuous temperature monitoring. Dehghani et al. (2021) emphasize that one of the most significant challenges in pharmaceutical cold chains is maintaining consistent temperature conditions across all stages of the supply chain. Their study identifies key issues such as temperature excursions, high operational costs, and compliance pressures. These challenges are further intensified in regions with extreme climates and long transportation distances, where maintaining stable conditions becomes more difficult. As a result, cold chain performance is not only a technical issue but also a strategic operational concern.

In recent years, technological advancements have played an increasingly important role in improving cold chain logistics. Ali et al. (2021) highlight the potential of Internet of Things (IoT) devices and blockchain technology in enhancing supply chain transparency and real-time monitoring. These technologies allow organizations to track temperature conditions continuously and respond quickly to any deviations, reducing the risk of product damage. In addition, digital systems improve coordination between supply chain actors, leading to more efficient and reliable distribution processes. This shift toward data-driven and technology-enabled logistics reflects a broader trend in supply chain management, where performance is closely linked to the ability to monitor and manage real-time information

While global studies provide valuable insights into cold chain management, research focusing on specific regional contexts remains limited. In the case of Saudi Arabia, supply chain challenges are influenced by unique environmental and operational factors. A recent study published in the *Saudi Pharmaceutical Journal* (2023) highlights that pharmaceutical supply chains in the Kingdom face issues such as poor demand forecasting, logistical inefficiencies, and supply disruptions. These factors contribute to medicine shortages and indicate broader weaknesses in supply chain performance. When considered alongside the country's high temperatures and long transportation routes, these challenges become even more critical for cold chain logistics, where maintaining product quality depends heavily on strict environmental control.

Despite the growing body of literature on cold chain logistics and pharmaceutical supply chains, several gaps remain evident. Much of the existing research focuses on general models or technological solutions without examining how these systems operate in real-world settings. In addition, many studies are concentrated in developed regions, with limited attention given to countries like Saudi Arabia, where environmental conditions and infrastructure create distinct challenges. Most importantly, there is a lack of empirical research that uses actual company data to assess cold chain performance, particularly within the pharmaceutical sector.

To address these gaps, this study examines cold chain logistics performance in Saudi Arabia using real-world case evidence. By focusing on practical applications rather than purely theoretical models, the research provides a more realistic understanding of how cold chain systems function in a challenging environment. It also contributes to the existing literature by offering context-specific insights and highlighting the importance of aligning technological solutions with operational realities in the pharmaceutical supply chain.

3. Methods

3.1 Research Design

This study adopts a qualitative multiple-case study approach to evaluate cold chain logistics performance in Saudi Arabia's pharmaceutical sector. A qualitative approach is appropriate because the research focuses on understanding

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real-world operational practices, decision-making processes, and organizational behavior rather than relying solely on numerical analysis.

3.2 Case Study Strategy
A case study strategy was selected to allow in-depth analysis of real-life logistics operations under temperature-controlled conditions. This method enables the exploration of complex variables such as environmental challenges, operational procedures, and human factors that influence cold chain performance.

3.3 Case Selection and Rationale
Two organizations were selected using purposive sampling:

Tamer Group: Represents the distribution and logistics stage of the pharmaceutical supply chain.
Medical Fakeeh: Represents the healthcare and end-user stage of the supply chain.

This selection allows comparison across different supply chain roles and provides a comprehensive understanding of cold chain performance.

3.4 Data Analysis Method
The collected data were analyzed using thematic analysis. Interview responses and operational insights were grouped into key themes such as temperature monitoring, operational challenges, and system effectiveness. A comparative analysis was then conducted to identify similarities and differences between the two organizations.

3.5 Reliability and Validity
To ensure reliability and validity, multiple data sources were used, including interviews and operational records. Triangulation helped confirm consistency across findings, while comparing two case organizations strengthened the robustness of the results.

3.6 Limitations
The study is limited to two case organizations, which may affect generalizability. Additionally, access to some internal data was restricted due to confidentiality constraints.

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4. Data Collection

4.1 Data Sources

The study relied on both primary and secondary data sources:

Primary data: Semi-structured interviews with professionals involved in supply chain and logistics operations.
Secondary data: Operational records, including temperature logs and product acceptance/rejection data.

4.2 Participants

Interview participants were selected based on their operational experience and managerial responsibilities in cold chain processes. The participants included:

- Engineer Alaa Alzamzami, Planning and Operations Manager at Tamer Logistics
- Dr. Ruba Barakat, Supply Chain Director at Medical Fakeeh
- Mr. Youssef Alzahrani, Warehouse Manager at Medical Fakeeh

4.3 Data Collection Methods

Semi-structured interviews were conducted using a set of guiding questions. This approach ensured consistency while allowing participants to provide detailed insights into operational practices. Follow-up questions were used to explore key issues in greater depth.

The interview questions focused on:

Cold chain workflow
Temperature monitoring technologies
Handling of temperature deviations
Operational challenges
Performance evaluation practices

4.4 Secondary Data Collection

Secondary data included temperature monitoring records and data logger outputs. These records were used to identify compliant and non-compliant shipments and to support interview findings with real operational evidence.

4.5 Data Collection Procedure

Data were collected over a defined period through scheduled interviews and review of company records. All data were organized and prepared for thematic analysis.

5. Results and Discussion

5.1 Findings from Tamer Group

This section presents the findings obtained from the case study of Tamer Group based on interview data.

5.1.1 Cold Chain Process

The cold chain process at Tamer Group begins with the storage of pharmaceutical products in specialized facilities equipped with precise temperature control systems, tailored to the requirements of each product type. Shipments are prepared within controlled environments to maintain temperature stability prior to dispatch.

Transportation is conducted using calibrated refrigerated trucks set to specific temperature ranges according to the nature of the products. During transit, continuous temperature monitoring is maintained to ensure that products remain within the required temperature range until final delivery.

5.1.2 Temperature Monitoring Technologies

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Tamer Group employs a combination of advanced monitoring technologies, including digital temperature sensors, GPS-enabled tracking systems, and data loggers placed within shipments.

These technologies enable:

- Real-time temperature monitoring
- Immediate alerts in case of temperature deviations
- Generation of complete temperature reports for the entire transportation journey

5.1.3 Handling Temperature Deviations

In the event of temperature deviations, a structured response protocol is implemented. The system generates an immediate alert, after which the transportation team communicates directly with the driver to identify the cause of the deviation, such as door openings or cooling system malfunctions.

If the issue persists, corrective actions are taken, including redirecting the vehicle to the nearest warehouse facility or replacing the vehicle when necessary.

5.1.4 Root Causes of Temperature Excursions

The most frequently identified causes of temperature deviations include:

- Prolonged opening of vehicle or storage doors
- Technical failures in refrigeration units
- Improper calibration of cooling systems

5.1.5 System Effectiveness

The cold chain system demonstrates high effectiveness, characterized by:

- Minimal temperature deviations
- Absence of reported product damage
- High levels of customer satisfaction
- Continuous monitoring by specialized teams

5.1.6 Key Challenges

The primary challenge identified is the impact of environmental conditions, particularly high temperatures during the summer season. These conditions increase the complexity of maintaining temperature control during loading and unloading operations, as well as during long-distance transportation between cities.

5.1.7 Cost and Quality Management

Tamer Group adopts a preventive maintenance strategy aimed at minimizing operational losses. The focus is placed on maintaining high-quality standards rather than reducing direct costs, ensuring the reliability of cold chain operations.

5.1.8 Key Factor Affecting Performance

Operational discipline is identified as the most critical factor influencing cold chain effectiveness. This includes adherence to procedures, minimizing human error, and ensuring consistent execution of operational protocols.

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5.2 Findings from Medical Fakeeh

This section presents the findings from Medical Fakeeh (Saudi Airlines Medical Services Company) , including both interview responses and data logger records.

5.2.1 Cold Chain Process

The cold chain process at Medical Fakeeh begins upon receiving temperature-sensitive pharmaceutical products from suppliers. During the receiving stage, temperature compliance is verified using data loggers to ensure that products remained within the required temperature range during transportation.

Products are then stored immediately in temperature-controlled warehouse environments. Based on internal demand, items are prepared and transported to hospital departments, such as pharmacies, using ice boxes and refrigerated vehicles. Temperature conditions are maintained throughout the process until final handover.

5.2.2 Temperature Monitoring Technologies

Medical Fakeeh utilizes data loggers to monitor incoming shipments and a dedicated warehouse temperature monitoring system to track storage conditions.

These systems provide:

- Continuous monitoring of temperature conditions
- Real-time alerts in case of deviations
- Accurate verification of temperature compliance

5.2.3 Handling Temperature Deviations

When a temperature excursion is detected through data logger readings, the shipment is rejected to ensure compliance with quality and safety standards.

5.2.4 Root Causes of Temperature Excursions

The main causes of temperature deviations include:

- Extended door openings
- Improper storage practices
- Equipment malfunctions (e.g., refrigerators or freezers)
- Power interruptions

To address these issues, the organization applies standard operating procedures, staff training, regular maintenance, and continuous monitoring systems.

5.2.5 System Effectiveness

The effectiveness of the cold chain system is evaluated using key performance indicators (KPIs), including:

- Temperature compliance rate
- On-time delivery performance
- Product loss percentage
- Audit results

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5.2.6 Cost and Quality Management

Cost efficiency and quality are balanced through route optimization, investment in reliable equipment, and the use of data analytics to minimize waste while maintaining compliance with required standards.

5.2.7 Key Factor Affecting Performance

The most significant factor affecting performance is the reliability of temperature control systems, supported by continuous monitoring and rapid response mechanisms.

5.2.8 Data Logger Results: Accepted and Rejected Products

The collected data indicates variations in product acceptance based on temperature compliance.

Table 1: Accepted products based on temperature compliance

Item Code	Product Name
16891	Humira 40 mg Injection
16924	Lantus SoloStar 100 units/ml
19640	Engerix-B Vaccine

Table 2: Rejected products due to temperature excursions

Item Code	Product Name
16060	Toujeo SoloStar 300
15519	Humalog Mix 25/75 KwikPen
16907	Wegovy 1.7

Table 3: Observed temperature ranges and shipment status

Data logger records indicate that acceptable temperature ranges are typically maintained between **2°C and 8°C**.

Category	Temperature Range	Status
Compliant	2°C – 8°C	Accepted
High Excursion	> 8°C	Rejected
Low Excursion	< 2°C	Rejected

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Figure 1: Temperature profile of a compliant pharmaceutical shipment recorded on 13/04/2026, showing stable conditions within the acceptable range (2°C–8°C), with temperatures fluctuating between approximately 4.2°C and 6.7°C throughout the monitoring period.

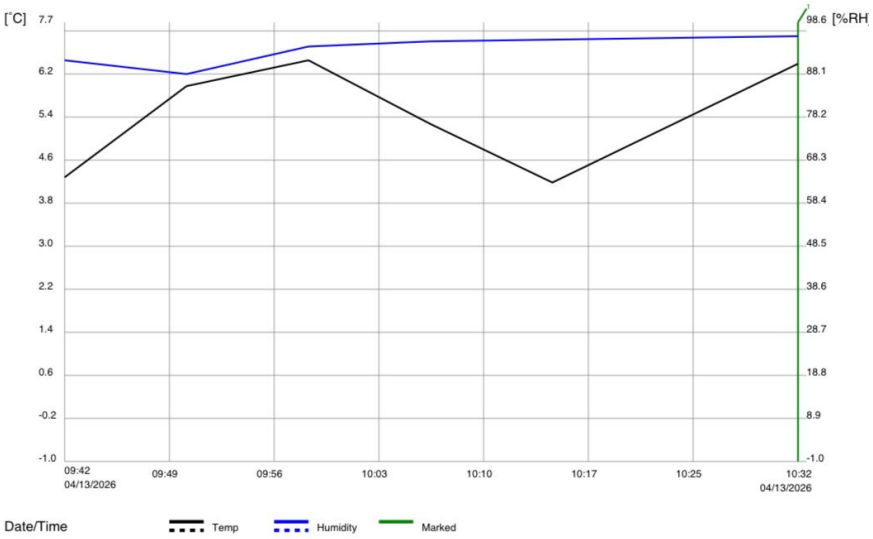
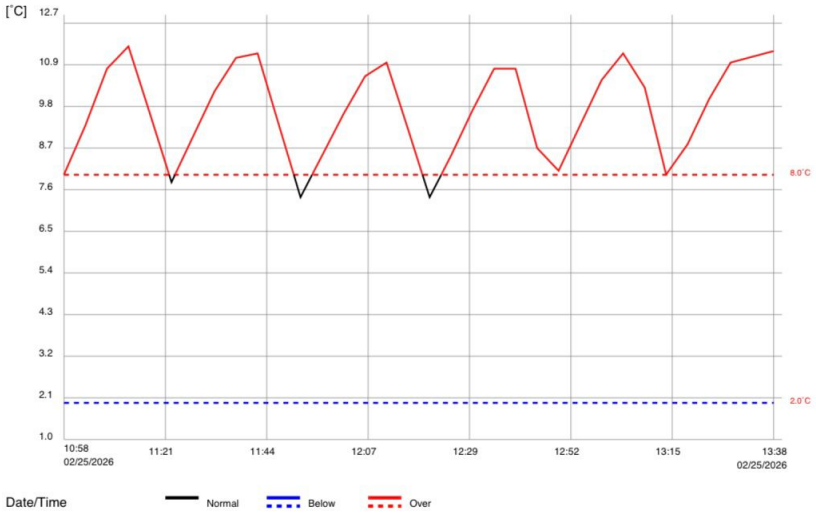
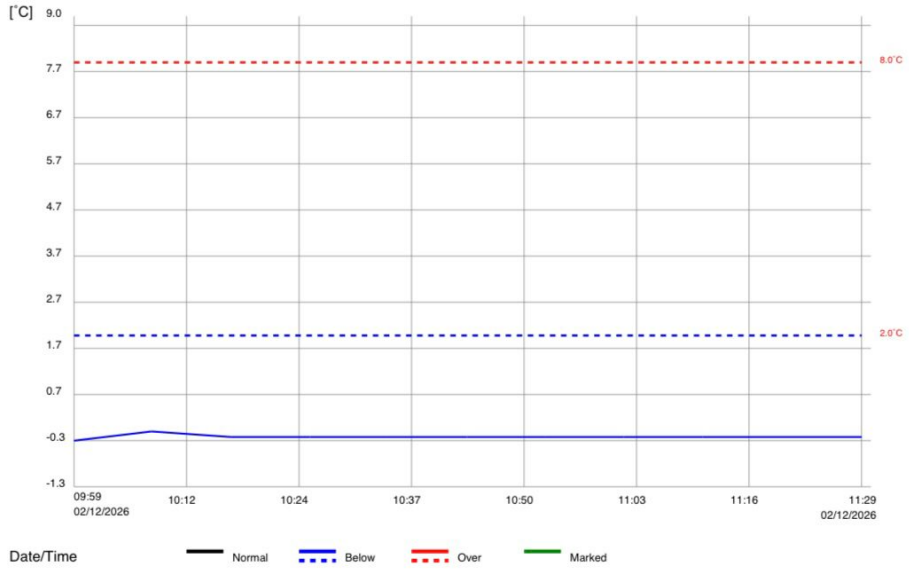


Figure 2: Temperature profile of a non-compliant shipment recorded on 25/02/2026, where temperatures exceeded the acceptable upper limit (8°C), reaching up to approximately 11.7°C, resulting in a temperature excursion and product rejection.



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Figure 3: Temperature profile of a non-compliant shipment recorded on 12/02/2026, where temperatures dropped below the acceptable lower limit (2°C), reaching approximately -0.3°C, indicating a temperature excursion and non-compliance.



Observed findings include:

- Stable temperature readings between approximately **4.0°C and 6.6°C** for compliant shipments
- Temperature excursions exceeding **8°C**, with recorded values up to **11.7°C**
- Instances of temperatures falling below **0°C**, indicating non-compliance

The recorded data demonstrates both compliant and non-compliant temperature conditions across different shipments, including instances of stability within the acceptable range and deviations above and below the defined limits.

Table 4: Comparative summary of cold chain logistics performance

A comparison of findings from both organizations is presented below:

Aspect	Tamer Group	Medical Fakeeh
Data Evidence	Interview-based findings	Interview + data logger records
Process initiation	Storage stage	Receiving stage
Transportation method	Refrigerated trucks	Ice boxes and refrigerated vehicles
Monitoring technologies	Sensors, GPS, data loggers	Data loggers and warehouse monitoring systems
Handling deviations	Corrective actions (rerouting, vehicle replacement)	Immediate rejection of shipments
Key challenge	Environmental conditions	Temperature control reliability
Key performance factor	Operational discipline	Monitoring and control reliability

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5.1 Interpretation of Cold Chain Performance

This research demonstrates that cold chain logistics are generally robust and efficient in Saudi Arabia's pharmaceutical industry, particularly in structured businesses like Tamer Group and Fakeeh Care Group. Both organizations have established clear protocols and monitoring systems and recognize the need of maintaining the proper temperature throughout the supply chain.This research demonstrates that cold chain logistics are generally robust and efficient in Saudi Arabia's pharmaceutical industry, particularly in structured businesses like Tamer Group and Fakeeh Care Group. Both organizations have established clear protocols and monitoring systems and recognize the need of maintaining the proper temperature throughout the supply chain.

But the two organizations have different ideas on what constitutes "effectiveness." Tamer Group uses a proactive, intervention-based strategy that continuously monitors and instantly corrects abnormalities. This illustrates a logistics-focused approach that places an emphasis on flexibility in handling disruptions and operational continuity. Medical Fakeeh, on the other hand, has a rigorous compliance-based strategy that ensures adherence to quality and safety requirements by immediately rejecting shipments that deviate.

These variations highlight two contrasting approaches to performance evaluation:

- Flexibility and operational efficiency (Tamer Group)
- Risk reduction and quality control (Medical Fakeeh)

When combined, these strategies show that cold chain efficacy is multifaceted, encompassing not only temperature control but also risk management, compliance assurance, and service continuity.

5.2 Comparing System Design and Operational Strengths

Tamer Group's solution, which uses cutting-edge technology like GPS tracking, real-time sensors, and continuous monitoring to maintain visibility across lengthy transportation routes, is more externally integrated and distribution-focused, according to a comparative analysis. This is in line with its function as a major distributor that operates in several areas, where control is essential while in route.

Medical Fakeeh's system, on the other hand, focuses on upholding stringent temperature conditions within hospital operations and is more internally managed and healthcare-focused. A systematic performance evaluation system in line with healthcare quality standards is reflected in the usage of KPIs like temperature compliance rate, product loss, and audit outcomes.

Organizational roles can be used to explain this difference:

- Distributors need to be adaptable and quick to react.
- Precision and compliance are essential for healthcare providers.

The results support the notion that cold chain logistics is not a one-size-fits-all system by indicating that, although both systems are functional, their design is influenced by organizational priorities and operational situations.

5.3 Technology's Contribution to Improving Cold Chain Performance

The findings firmly corroborate the claims made in the literature review regarding the vital role that technology plays in enhancing cold chain performance. Both companies use digital sensors and data loggers for temperature monitoring, which allows for continuous surveillance and quick reaction to abnormalities.

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In Tamer Group's situation, dynamic intervention during transportation is made possible by the integration of GPS tracking with real-time monitoring systems, which lowers the possibility of product loss. In a similar vein, Medical Fakeeh makes sure that only compatible products are admitted into the system by relying on data logger verification and warehouse monitoring technologies.

These results are consistent with research like that conducted by Ali et al. (2021), which highlights the significance of IoT and digital technologies in improving supply chain responsiveness and visibility.

But the results also show that technology by itself cannot ensure efficacy. Medical Fakeeh emphasized system reliability, while Tamer Group determined that operational discipline was the most important issue. This suggests that human aspects, including following protocols, treating things correctly, and keeping an eye on things consistently, are just as crucial to guaranteeing system performance.

5.4 Saudi Arabia's Operational and Environmental Challenges

The important influence of environmental factors, especially the intense heat in Saudi Arabia, is a major finding in both case studies. Maintaining stable conditions becomes more challenging in high ambient temperatures, particularly during:

- Operations related to loading and unloading
- Transportation over long distances between cities

These results are in line with research that highlights climate conditions as a significant obstacle to cold chain logistics, especially in areas with severe weather. Along with environmental difficulties, operational problems like:

- Door openings
- Equipment failures

Temperature excursions were found to be frequently caused by power outages. These elements demonstrate the intricacy of cold chain management, where performance is influenced by both internal operating procedures and external environmental influences.

Medical Fakeeh's inclusion of both complying and non-compliant temperature data highlights the fact that even well-designed systems are susceptible to interruptions, underscoring the necessity of ongoing observation and development.

5.5 Cost Efficiency and Quality Trade-Off

The results show that both companies place a higher priority on product safety and quality than on cutting costs, which is crucial in the pharmaceutical industry. While Medical Fakeeh employs data-driven decision-making and route optimization to increase efficiency without sacrificing standards, Tamer Group concentrates on preventive maintenance to lower long-term operational hazards.

This validates current supply chain theories that emphasize the significance of striking a balance between service quality and cost effectiveness. The potential cost advantages in pharmaceutical logistics are greatly outweighed by the consequences of failure, such as product degradation or patient health risks.

5.6 Research Restrictions

Although the study offers insightful real-world information, a number of limitations should be noted:

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- Only two case study organizations served as the basis for the study, which restricts how broadly the results may be applied.
- Tamer Group's data is mostly qualitative (based on interviews), with no quantitative data to back it up.
- The study concentrates on particular operational settings, which could not accurately reflect entire Saudi Arabian pharmaceutical supply chains.
- Due to the short data collection period, seasonal fluctuations in performance might not be captured.

By using a bigger sample of organizations, integrating qualitative and quantitative data, and carrying out longitudinal studies to track performance over time, future research could overcome these constraints.

6. Conclusion

Using case studies from Tamer Group and Medical Fakeeh, this study investigated the efficacy of cold chain logistics techniques in Saudi Arabia's pharmaceutical industry.

The results show that the Kingdom's cold chain systems are often strong, organized, and backed by contemporary technology, enabling businesses to preserve temperature integrity and guarantee product safety. However, a number of variables, such as technology, operational discipline, system design, and environmental circumstances, affect efficacy.

When the two organizations are compared, it becomes clear that although they both attain high performance levels, their methods differ:

- Tamer Group prioritizes operational flexibility and real-time monitoring.
- Medical Fakeeh places a strong emphasis on quality assurance and rigorous compliance.

Despite these advantages, the performance of the cold chain is nevertheless impacted by human factors, operational disturbances, and harsh climate conditions.

Overall, the study comes to the conclusion that, even if current procedures work well, ongoing development is required to boost system resilience, lower risks, and guarantee the safe delivery of pharmaceutical items.

7. Recommendations

7.1 Recommendations for Tamer Group

The results show that Tamer Group has excellent operational management and real-time monitoring. Nonetheless, the following areas could want improvement:

7.1.1 Strengthening Temperature Deviation Preventive Measures

Even when corrective measures work, the business should concentrate more on lowering the frequency of deviations by:

- Enhancing door control systems for loading and unloading
- Improving refrigeration unit calibration schedules
- Stricter loading procedures are being implemented

This lessens the need for reactive solutions.

7.1.2 Improving Operational Discipline (Human Factor Control)

Given that the crucial element was found to be operational discipline:

- Organize frequent staff training initiatives

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- Install systems that track drivers' and handlers' performance.
- Make use of automatic compliance warnings and checklists

This reduces human mistake, which is still a significant danger.

7.1.3 Growing Systems for Predictive Monitoring

Real-time tracking is already used by Tamer; however it might be improved by:

- Predictive analytics (AI/IoT) for failure prediction
- Risk detection prior to temperature fluctuations

Transitions from a reactive to a proactive to a predictive system.

7.2 Recommendations for Fakeeh Care Group

Medical Fakeeh is more reactive (rejecting shipments), yet it exhibits good compliance and quality control. Enhancements may consist of:

7.2.1 Introducing Transportation-Related Real-Time Monitoring

Currently, post-delivery data recorders play a major role in decision-making. They should:

- Use real-time temperature monitoring while in transit
- Encourage early intervention rather than rejection

This lowers the cost and waste of products.

7.2.2 Using Early Intervention to Reduce Product Rejection

Rather than simply rejecting shipments that do not comply:

- During transportation, implement corrective action procedures.
- Coordinate with logistics companies and suppliers.

to increase productivity without sacrificing quality.

7.2.3 Improving Operational Risk Backup Systems

Given that equipment malfunctions and power outages are among the causes:

- Install generators or other backup power systems.
- Make use of redundant cooling systems
- Boost the frequency of maintenance

This increases the dependability of the system.

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Biography

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Leen Alabsi is an undergraduate student in Operations and Supply Chain Management at Effat University in Jeddah, Saudi Arabia, with a strong academic focus on healthcare and pharmaceutical supply chains, particularly cold chain logistics and the factors influencing performance, reliability, and quality in temperature-sensitive distribution systems. She completed her internship at Medical Fakeeh (Saudi Airlines Medical Services Company), where she gained practical exposure to healthcare operations and supply chain processes within a clinical environment, developing a deeper understanding of real-world challenges in managing medical and pharmaceutical flows, including the critical importance of temperature control and operational efficiency.

Manar Sharaf Aldeen is an undergraduate student in Operations and Supply Chain Management at Effat University, based in Jeddah. She has a strong academic interest in healthcare and pharmaceutical supply chains, particularly in cold chain logistics and the management of temperature-sensitive products. Her studies focus on improving operational efficiency, reliability, and overall supply chain performance. Manar completed an internship with Nestlé, where she participated in a sales and development program that provided practical exposure to real-world market operations. During this experience, she worked closely with BinDawood for a week, gaining insight into retail execution, product placement strategies, and customer purchasing behavior. She also took part in warehouse visits at Tamer Logistics, where she observed key logistics functions such as inventory management, storage systems, and distribution processes. These experiences enhanced her understanding of supply chain coordination and operational flow. Additionally, through her engagement with Tamer Group, Manar gained valuable insight into pharmaceutical distribution and supply chain management. She developed a deeper understanding of regulatory requirements, temperature control, and the importance of maintaining product integrity in medical supply chains.Manar continues to build her analytical and problem-solving skills, with a strong interest in leveraging technology and data-driven solutions to improve supply chain efficiency. She aims to contribute to the development of reliable and sustainable healthcare supply chains in the future.

Ghalia AlHazmi is an undergraduate student in Operations and Supply Chain Management at Effat University in Jeddah, Saudi Arabia. Her academic interests focus on logistics management, process optimization, and supply chain coordination, particularly within healthcare and service-oriented industries. Throughout her studies, she has developed strong analytical and problem-solving skills through coursework and collaborative projects that required evaluating real-world supply chain challenges and proposing practical, data-driven solutions.Ghalia has actively contributed to research projects related to cold chain logistics, where she participated in analyzing operational processes, identifying performance gaps, and examining the impact of environmental and human factors on supply chain efficiency. She is particularly interested in improving operational reliability and ensuring consistent performance in temperature-sensitive distribution systems. Her work reflects a strong attention to detail and an ability to connect theoretical knowledge with practical applications. She is also experienced in teamwork and project coordination, having worked in diverse groups where effective communication and role clarity were essential for achieving project objectives. Ghalia aims to pursue a career in supply chain management, where she can contribute to enhancing logistics performance and supporting efficient, high-quality service delivery in complex environments.

Talah Alzuabi is an undergraduate student in Operations and Supply Chain Management at Effat University in Jeddah, Saudi Arabia. Her academic focus includes supply chain strategy, logistics planning, and operational performance improvement. She has a strong interest in understanding how strategic decisions impact efficiency, cost management, and overall supply chain effectiveness across different industries. Throughout her academic journey, Talah has participated in several team-based projects that involved analyzing real-life case studies, including healthcare and pharmaceutical supply chains. She has developed the ability to evaluate operational challenges, assess trade-offs between cost, time, and quality, and propose structured solutions to improve system performance. Her involvement in research on cold chain logistics has further strengthened her understanding of temperature-controlled systems and the importance of maintaining product integrity under challenging conditions.Talah is particularly interested in decision-making processes within supply chains and how organizations can balance competing priorities to achieve optimal outcomes. She is known for her organizational skills, adaptability, and ability to contribute effectively within team environments. In the future, she aims to build a career in supply chain and logistics management, focusing on improving operational efficiency and supporting sustainable and resilient supply chain systems.

Yasmina Sagga is an undergraduate student in Operations and Supply Chain Management at Effat University in Jeddah, Saudi Arabia. Her academic interests focus on logistics systems, operational efficiency, and data-driven

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decision-making, particularly within healthcare and service supply chains. She is passionate about understanding how supply chain processes can be optimized to improve performance, reduce inefficiencies, and ensure reliable service delivery in complex environments. Throughout her studies, Yasmina has developed strong analytical and research skills through coursework and collaborative projects that involve evaluating real-world supply chain challenges. She has contributed to projects focused on cold chain logistics, where she participated in analyzing system performance, identifying operational risks, and assessing the effectiveness of temperature monitoring practices. This experience has strengthened her ability to connect theoretical concepts with practical applications in the pharmaceutical supply chain. Yasmina is particularly interested in the role of data and technology in improving supply chain visibility and decision-making. She values structured problem-solving and effective communication, which has enabled her to work efficiently within team-based environments. Known for her attention to detail and commitment to quality, she aims to pursue a career in supply chain management where she can contribute to enhancing logistics performance and supporting efficient, reliable, and sustainable operations.

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